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PROGRAM NAME : EX_SDTM.SAS

PROJECT : SDTM EXPOSURE DOMAIN DEVELOPMENT

STUDY : 009/01

AUTHOR : NITHISH M S

DATE CREATED : 26MAR2026

PLATFORM : SAS 9.4

DESCRIPTION:

THIS PROGRAM CREATES THE SDTM EXPOSURE (EX) DOMAIN DATASET FROM RAW DATA.

IT INCLUDES DERIVATION OF KEY VARIABLES SUCH AS:

- USUBJID (UNIQUE SUBJECT IDENTIFIER)
- EXSTDTC / EXENDTC (ISO8601 DATETIME)
- EXSTDY / EXENDY (STUDY DAY VARIABLES)
- EXDUR (DURATION OF EXPOSURE)
- EXSEQ (SEQUENCE NUMBER WITHIN SUBJECT)

THE PROGRAM INTEGRATES EX AND DM DATASETS TO CALCULATE STUDY DAY VARIABLES

BASED ON REFERENCE START DATE (RFSTDN).

INPUT DATASETS:

- RAW EXPOSURE DATASET (EXCEL)
- RAW DEMOGRAPHICS DATASET (EXCEL)

OUTPUT DATASETS:

- **FINAL_EX** : SDTM-COMPLIANT EXPOSURE DATASET
- **EX_FINAL** : FINAL LABELED DATASET
- **XPT FILE** : EXPOSURE.XPT (SUBMISSION-READY)

MODIFICATION HISTORY:

DATE **AUTHOR** **DESCRIPTION**

27MAR2026 NITHISH INITIAL VERSION

/* RAW DATA IMPORT */

PROC IMPORT DATAFILE="/HOME/U64168920/PROJECT SDTM EXPOSURE RAW
DATAFILE.XLSX"

OUT=EX

DBMS=XLSX REPLACE;

RUN;

Table: WORK.EX		View: Column names		Filter: (none)									
Total rows: 41		Total columns: 14										Rows 1-41	
	VISIT	DSDT	DSDTM	DOSTP	DOSE	DOSEN	TRT	SITE	STUDY	SUBJID	ROUTE	FREQ	FAST
1	Period-1	10/22/2018	9:00:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S1	oral	once	yes
2	Period-2	11/22/2018	9:00:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S1	oral	once	yes
3	Period-1	10/22/2018	9:02:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S2	oral	once	yes
4	Period-2	11/22/2018	9:02:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S2	oral	once	yes
5	Period-1	10/22/2018	9:06:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S4	oral	once	yes
6	Period-1	10/22/2018	9:08:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S5	oral	once	yes
7	Period-2	11/22/2018	9:08:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S5	oral	once	yes
8	Period-1	10/22/2018	9:10:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S6	oral	once	yes
9	Period-2	11/22/2018	9:10:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S6	oral	once	yes
10	Period-1	10/22/2018	9:12:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S7	oral	once	yes
11	Period-2	11/22/2018	9:12:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S7	oral	once	yes
12	Period-1	10/22/2018	9:14:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S8	oral	once	yes
13	Period-2	11/22/2018	9:14:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S8	oral	once	yes
14	Period-1	10/22/2018	9:16:00	TABLET	10 mg	Oxybutynin	REF	1	009/01	S9	oral	once	yes
15	Period-2	11/22/2018	9:16:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S9	oral	once	yes
16	Period-1	10/22/2018	9:18:00	TABLET	10 mg	Oxybutynin	TES	1	009/01	S10	oral	once	yes

DATA EX1;

SET EX;

STUDYID=STRIP(STUDY);

DOMAIN="EX";

SITEID=STRIP(SITE);

SUBJECT=STRIP(SUBJID);

USUBJID=STRIP(STUDYID)||"-"||STRIP(SITEID)||"-"||STRIP(SUBJID);

EXTRT=STRIP(TRT);

EXDOSE=INPUT(SUBSTR(DOSE,1,2),BEST.);

EXDOSU=SUBSTR(DOSE,3,4);

EXDOSTXT=DOSEN;

EXDOSFRQ=UPCASE(FREQ);

EXROUTE=UPCASE(ROUTE);

EXFAST=UPCASE(SUBSTR(FAST,1,1));

DSDTN=INPUT(DSDT,MMDDYY10.);

EXSTDTC=PUT(DSDTN,IS8601DA.)||"T"||PUT(DSDTM, TOD8.);

EXSTDN=DATEPART(INPUT(EXSTDTC,IS8601DT.));

FORMAT EXSTDN DATE9.;

EXENDTC=PUT(DSDTN,IS8601DA.)||"T"||PUT(DSDTM,TOD8.);

EXENDN=DATEPART(INPUT(EXENDTC,IS8601DT.));

FORMAT EXENDN DATE9.;

RUN;

Table: WORK.EXT		View: Column names		Filter: (none)																									
Total rows: 41		Total columns: 31																											
VISIT	DSOT	DSOTM	DOSTP	DOSE	DOSEN	TRT	SITE	STUDY	SUBJID	ROUTE	FREQ	FAST	N	STUDYID	DOMAIN	SITEID	SUBJECT	USUBJID	EXTRT	EXDOSE	EXDOSU	EXDOSTX	EXDOSFRG	EXROUTE	EXFAST	DSOTN	EXSTOTC	EXSTDTN	EXENDTC
1	Period-1	10/22/2018	9:00:00	TABLET	10 mg	Oral	REF	1	009/01	S1	oral	once	yes	009/01	EX	1	S1	009/01-1.51	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:00:00	22OCT2018	2018-10-22T09:00:00
2	Period-2	11/22/2018	9:00:00	TABLET	10 mg	Oral	REF	1	009/01	S1	oral	once	yes	009/01	EX	1	S1	009/01-1.51	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:00:00	22NOV2018	2018-11-22T09:00:00
3	Period-1	10/22/2018	9:02:00	TABLET	10 mg	Oral	REF	1	009/01	S2	oral	once	yes	009/01	EX	1	S2	009/01-1.52	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:02:00	22OCT2018	2018-10-22T09:02:00
4	Period-2	11/22/2018	9:02:00	TABLET	10 mg	Oral	REF	1	009/01	S2	oral	once	yes	009/01	EX	1	S2	009/01-1.52	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:02:00	22NOV2018	2018-11-22T09:02:00
5	Period-1	10/22/2018	9:06:00	TABLET	10 mg	Oral	REF	1	009/01	S4	oral	once	yes	009/01	EX	1	S4	009/01-1.54	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:06:00	22OCT2018	2018-10-22T09:06:00
6	Period-1	10/22/2018	9:08:00	TABLET	10 mg	Oral	REF	1	009/01	S5	oral	once	yes	009/01	EX	1	S5	009/01-1.55	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:08:00	22OCT2018	2018-10-22T09:08:00
7	Period-2	11/22/2018	9:08:00	TABLET	10 mg	Oral	REF	1	009/01	S5	oral	once	yes	009/01	EX	1	S5	009/01-1.55	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:08:00	22NOV2018	2018-11-22T09:08:00
8	Period-1	10/22/2018	9:10:00	TABLET	10 mg	Oral	REF	1	009/01	S6	oral	once	yes	009/01	EX	1	S6	009/01-1.56	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:10:00	22OCT2018	2018-10-22T09:10:00
9	Period-2	11/22/2018	9:10:00	TABLET	10 mg	Oral	REF	1	009/01	S6	oral	once	yes	009/01	EX	1	S6	009/01-1.56	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:10:00	22NOV2018	2018-11-22T09:10:00
10	Period-1	10/22/2018	9:12:00	TABLET	10 mg	Oral	REF	1	009/01	S7	oral	once	yes	009/01	EX	1	S7	009/01-1.57	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:12:00	22OCT2018	2018-10-22T09:12:00
11	Period-2	11/22/2018	9:12:00	TABLET	10 mg	Oral	REF	1	009/01	S7	oral	once	yes	009/01	EX	1	S7	009/01-1.57	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:12:00	22NOV2018	2018-11-22T09:12:00
12	Period-1	10/22/2018	9:14:00	TABLET	10 mg	Oral	REF	1	009/01	S8	oral	once	yes	009/01	EX	1	S8	009/01-1.58	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:14:00	22OCT2018	2018-10-22T09:14:00
13	Period-2	11/22/2018	9:14:00	TABLET	10 mg	Oral	REF	1	009/01	S8	oral	once	yes	009/01	EX	1	S8	009/01-1.58	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:14:00	22NOV2018	2018-11-22T09:14:00
14	Period-1	10/22/2018	9:16:00	TABLET	10 mg	Oral	REF	1	009/01	S9	oral	once	yes	009/01	EX	1	S9	009/01-1.59	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:16:00	22OCT2018	2018-10-22T09:16:00
15	Period-2	11/22/2018	9:16:00	TABLET	10 mg	Oral	REF	1	009/01	S9	oral	once	yes	009/01	EX	1	S9	009/01-1.59	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:16:00	22NOV2018	2018-11-22T09:16:00
16	Period-1	10/22/2018	9:18:00	TABLET	10 mg	Oral	REF	1	009/01	S10	oral	once	yes	009/01	EX	1	S10	009/01-1.510	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:18:00	22OCT2018	2018-10-22T09:18:00
17	Period-2	10/22/2018	9:20:00	TABLET	10 mg	Oral	REF	1	009/01	S11	oral	once	yes	009/01	EX	1	S11	009/01-1.511	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:20:00	22OCT2018	2018-10-22T09:20:00
18	Period-2	11/22/2018	9:20:00	TABLET	10 mg	Oral	REF	1	009/01	S11	oral	once	yes	009/01	EX	1	S11	009/01-1.511	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:20:00	22NOV2018	2018-11-22T09:20:00
19	Period-1	10/22/2018	9:22:00	TABLET	10 mg	Oral	REF	1	009/01	S12	oral	once	yes	009/01	EX	1	S12	009/01-1.512	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:22:00	22OCT2018	2018-10-22T09:22:00
20	Period-2	11/22/2018	9:22:00	TABLET	10 mg	Oral	REF	1	009/01	S12	oral	once	yes	009/01	EX	1	S12	009/01-1.512	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:22:00	22NOV2018	2018-11-22T09:22:00
21	Period-1	10/22/2018	9:24:00	TABLET	10 mg	Oral	REF	1	009/01	S13	oral	once	yes	009/01	EX	1	S13	009/01-1.513	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:24:00	22OCT2018	2018-10-22T09:24:00
22	Period-2	11/22/2018	9:24:00	TABLET	10 mg	Oral	REF	1	009/01	S13	oral	once	yes	009/01	EX	1	S13	009/01-1.513	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:24:00	22NOV2018	2018-11-22T09:24:00
23	Period-1	10/22/2018	9:26:00	TABLET	10 mg	Oral	REF	1	009/01	S14	oral	once	yes	009/01	EX	1	S14	009/01-1.514	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:26:00	22OCT2018	2018-10-22T09:26:00
24	Period-2	11/22/2018	9:26:00	TABLET	10 mg	Oral	REF	1	009/01	S14	oral	once	yes	009/01	EX	1	S14	009/01-1.514	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:26:00	22NOV2018	2018-11-22T09:26:00
25	Period-1	10/22/2018	9:28:00	TABLET	10 mg	Oral	REF	1	009/01	S15	oral	once	yes	009/01	EX	1	S15	009/01-1.515	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:28:00	22OCT2018	2018-10-22T09:28:00
26	Period-2	11/22/2018	9:28:00	TABLET	10 mg	Oral	REF	1	009/01	S15	oral	once	yes	009/01	EX	1	S15	009/01-1.515	REF	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:28:00	22NOV2018	2018-11-22T09:28:00
27	Period-1	10/22/2018	9:30:00	TABLET	10 mg	Oral	REF	1	009/01	S16	oral	once	yes	009/01	EX	1	S16	009/01-1.516	REF	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:30:00	22OCT2018	2018-10-22T09:30:00
28	Period-2	11/22/2018	9:30:00	TABLET	10 mg	Oral	REF	1	009/01	S16	oral	once	yes	009/01	EX	1	S16	009/01-1.516	TES	10	mg	Oral	ONCE	ORAL	Y	21510	2018-11-22T09:30:00	22NOV2018	2018-11-22T09:30:00
29	Period-1	10/22/2018	9:32:00	TABLET	10 mg	Oral	REF	1	009/01	S17	oral	once	yes	009/01	EX	1	S17	009/01-1.517	TES	10	mg	Oral	ONCE	ORAL	Y	21479	2018-10-22T09:32:00	22OCT2018	2018-10-22T09:32:00

PROC IMPORT DATAFILE="/HOME/U64168920/PROJECT SDTM DM RAW
DATAFILE.XLSX"

OUT=DM

DBMS=XLSX REPLACE;

GETNAMES=YES;

QUIT;

Table: WORK.DM		View: Column names		    Filter: (none)															
Total rows: 55		Total columns: 23																	
SUBJID	VISIT	GEN	AGEUN	ETH	ETHO	HT	HTU	WT	WTU	BMI	BMU	ENTDT	ENRTM	CMPDT	CMPTM	infdt	infm	inv	SITE
S1	SCREENING	Male	20	Asian		175	CM	59	KG	19.3	KG/M2	10/21/2018	20:00:00	24NOV2023	9:00:00	24OCT2018	18:00:00	Dr.A.Sumarlata	1
S2	SCREENING	Female	25	Asian		150	CM	89	KG	24.9	KG/M2	10/21/2018	18:00:00	24NOV2023	9:00:00	21OCT2018	20:00:00	Dr.A.Sumarlata	1
S3	SCREENING	Female	24	Asian		140	CM	78	KG	24.9	KG/M2	10/21/2018	19:00:00	24NOV2023	9:00:00	24OCT2018	20:00:00	Dr.A.Sumarlata	1
S4	SCREENING	Male	24	Asian		170	CM	68	KG	22.9	KG/M2	10/21/2018	18:00:00	24NOV2023	9:00:00	22OCT2018	18:00:00	Dr.A.Sumarlata	1
S5	SCREENING	Male	21	Other		172	CM	60	KG	19.8	KG/M2	10/21/2018	20:00:00	24NOV2023	9:00:00	21OCT2018	18:00:00	Dr.A.Sumarlata	1
S6	SCREENING	Male	22	Asian		170	CM	52	KG	19.1	KG/M2	10/21/2018	18:00:00	24NOV2023	9:00:00	24OCT2018	18:00:00	Dr.A.Sumarlata	1
S7	SCREENING	Male	24	Asian		170	CM	65	KG	23.4	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	22OCT2018	18:00:00	Dr.A.Sumarlata	1
S8	SCREENING	Male	25	Asian		155	CM	72	KG	26.3	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	22OCT2018	18:00:00	Dr.A.Sumarlata	1
S9	SCREENING	Female	28	Asian		162	CM	68	KG	20.6	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	23OCT2018	18:00:00	Dr.A.Sumarlata	1
S10	SCREENING	Female	31	Asian	WHITE	167	CM	60	KG	21.5	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	23OCT2018	18:00:00	Dr.A.Sumarlata	1
S11	SCREENING	Female	27	Asian		154	CM	65	KG	19.5	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	25OCT2018	18:00:00	Dr.A.Sumarlata	1
S12	SCREENING	Female	29	Asian		150	CM	62	KG	20.7	KG/M2	10/22/2018	18:00:00	24NOV2023	9:00:00	24OCT2018	18:00:00	Dr.A.Sumarlata	1

DATA DM1;

SET DM;

STUDYID=STRIP(STUDY);

DOMAIN="DM";

SITEID=STRIP(SITE);

SUBJECT=STRIP(SUBJID);

USUBJID=STRIP(STUDYID)||"-"||STRIP(SITEID)||"-"||STRIP(SUBJID);

ENRDTN=INPUT(ENTDT,MMDDYY10.);

RFSTDTC=PUT(ENRDTN,IS8601DA.)||"T"||PUT(ENRTM,TOD8.);

RFSTDN=DATEPART(INPUT(RFSTDTC,IS8601DT.));

KEEP USUBJID RFSTDN RFSTDTC;

FORMAT RFSTDN DATE9.;

RUN;

OUTPUT DATA

View: Column names ▼

Total rows: 55 Total columns: 3

	USUBJID	RFSTDTC	RFSTDTN
1	009/01-1-S1	2018-10-21T20:00:00	21OCT2018
2	009/01-1-S2	2018-10-21T18:00:00	21OCT2018
3	009/01-1-S3	2018-10-21T19:00:00	21OCT2018
4	009/01-1-S4	2018-10-21T18:00:00	21OCT2018
5	009/01-1-S5	2018-10-21T20:00:00	21OCT2018
6	009/01-1-S6	2018-10-21T18:00:00	21OCT2018

```
PROC SORT DATA=EX1 OUT=EX3;BY USUBJID;RUN;
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```
PROC SORT DATA=DM1 OUT=DM3;BY USUBJID;RUN;
```

DATA DM_EX;

```

MERGE DM3(IN=A) EX3(IN=B);

```

BY USUBJID;

IF A AND B;

RUN;

Total rows: 41. Total columns: 33		WORK_DM_EX - View		Column names -		    Filter: (none)																					
		INSTCTN	VISIT	DOSTDT	DOSMT	DOSTP	DOSE	DOSEN	TRT	SITE STUDY	SUBJID	ROUTE	FREQ	FAST	N	STUDYID	DORMAN	SITSD	SUBJECT	REF	EDDOSE	EXDOSDT	EXDOSPRG	EXROUTE	EXFAST	DSOINT	EXSTDTC
		USUBJID	HSTSTDTC																								
1	20001-1-51	1016-10-217220.000	210CT2016	Period 1	10/22/2018 9:00:00	TABLET	10 mg	Oxycodone	YES	1	00001	S1	oral	once	yes	00001	EX	1	S1	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:00:00
2	20001-1-51	1016-10-217220.000	210CT2016	Period 2	11/22/2018 9:00:00	TABLET	10 mg	Oxycodone	YES	1	00001	S1	oral	once	yes	00001	EX	1	S1	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2010-11-22T09:00:00
3	20001-1-510	1016-10-221218.000	220CT2016	Period 1	10/22/2018 9:18:00	TABLET	10 mg	Oxycodone	YES	1	00001	S10	oral	once	yes	00001	EX	1	S10	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:18:00
4	20001-1-511	1016-10-22118.000	220CT2016	Period 1	10/22/2018 9:20:00	TABLET	10 mg	Oxycodone	YES	1	00001	S11	oral	once	yes	00001	EX	1	S11	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:20:00
5	20001-1-511	1016-10-22118.000	220CT2016	Period 2	11/22/2018 9:20:00	TABLET	10 mg	Oxycodone	YES	1	00001	S11	oral	once	yes	00001	EX	1	S11	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2010-11-22T09:20:00
6	20001-1-512	1016-10-22118.000	220CT2016	Period 1	10/22/2018 9:22:00	TABLET	10 mg	Oxycodone	YES	1	00001	S12	oral	once	yes	00001	EX	1	S12	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:22:00
7	20001-1-512	1016-10-22118.000	220CT2016	Period 2	11/22/2018 9:22:00	TABLET	10 mg	Oxycodone	YES	1	00001	S12	oral	once	yes	00001	EX	1	S12	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2010-11-22T09:22:00
8	20001-1-513	1016-10-22118.000	220CT2016	Period 1	10/22/2018 9:24:00	TABLET	10 mg	Oxycodone	YES	1	00001	S13	oral	once	yes	00001	EX	1	S13	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:24:00
9	20001-1-513	1016-10-22118.000	220CT2016	Period 2	11/22/2018 9:24:00	TABLET	10 mg	Oxycodone	YES	1	00001	S13	oral	once	yes	00001	EX	1	S13	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-11-22T09:24:00
10	20001-1-514	1016-10-22118.000	220CT2016	Period 1	10/22/2018 9:26:00	TABLET	10 mg	Oxycodone	YES	1	00001	S14	oral	once	yes	00001	EX	1	S14	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:26:00
11	20001-1-514	1016-10-22118.000	220CT2016	Period 2	11/22/2018 9:26:00	TABLET	10 mg	Oxycodone	YES	1	00001	S14	oral	once	yes	00001	EX	1	S14	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2010-11-22T09:26:00
12	20001-1-515	1016-10-22118.000	220CT2016	Period 1	10/22/2018 9:28:00	TABLET	10 mg	Oxycodone	YES	1	00001	S15	oral	once	yes	00001	EX	1	S15	TES	10 mg	Oxycodone	ONCE	ORAL	Y	21478	2010-10-22T09:28:00

DATA EX2;

```
SET DM_EX;
```

IF EXSTDN > . AND RFSTDN > . THEN DO;

IF EXSTDTN>=RFSTDTN THEN EXSTDY=EXSTDTN-RFSTDTN+1;

```
ELSE IF EXSTDY<RFSTDY THEN EXSTDY=EXSTDY-RFSTDY;
```

END;

```

IF EXENDTN > . AND RFSTDTN > . THEN DO;
IF EXENDTN >= RFSTDTN THEN EXENDY = EXENDTN - RFSTDTN + 1;
ELSE IF EXENDTN < RFSTDTN THEN EXENDY = EXENDTN - RFSTDTN;
END;

```

```

EXDUR1=EXENDY-EXSTDY+1;
EXDUR=("P"||PUT(EXDUR1,BEST.)||"D");
EXDUR=COMPRESS(EXDUR);
EXDOSFRM=STRIP(DOSTP);
RUN;

```

CODE LOG RESULTS OUTPUT DATA

Table: WORKEX2 - View Column names Filter: (none)

Total rows: 41 Total columns: 38

	USUBJID	RFSTDTN	RFSTDTN	VISIT	DOST	DOSTM	DOSTP	DOSE	DOSEN	TRT	SITE	STUDY	SUBJID	ROUTE	FREQ	FAST	N	STUDYID	DOMAIN	SITEID	SUBJECT	EXTRT	EXDOSE	EXDOSU	EXDOSTXT	EXDOSFRQ	EXROUTE	EXFAST	DSDTN	EXSTDTC
1	00901-1.51	2018-10-21T20:00:00	21OCT2018	Period-1	10/22/2018	9:00:00	TABLET	10 mg	Oxycodone	REF	1	00901	S1	oral	once	yes		00901	EX	1	S1	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-22T09:00:00	
2	00901-1.51	2018-10-21T20:00:00	21OCT2018	Period-2	11/23/2018	9:00:00	TABLET	10 mg	Oxycodone	YES	1	00901	S1	oral	once	yes		00901	EX	1	S1	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-22T09:00:00	
3	00901-1.510	2018-10-22T18:00:00	22OCT2018	Period-1	10/23/2018	9:18:00	TABLET	10 mg	Oxycodone	YES	1	00901	S10	oral	once	yes		00901	EX	1	S10	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-22T09:18:00	
4	00901-1.511	2018-10-23T18:00:00	23OCT2018	Period-1	10/23/2018	9:20:00	TABLET	10 mg	Oxycodone	YES	1	00901	S11	oral	once	yes		00901	EX	1	S11	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-23T09:20:00	
5	00901-1.511	2018-10-23T18:00:00	23OCT2018	Period-2	11/23/2018	9:20:00	TABLET	10 mg	Oxycodone	REF	1	00901	S11	oral	once	yes		00901	EX	1	S11	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-23T09:20:00	
6	00901-1.512	2018-10-23T18:00:00	23OCT2018	Period-1	10/22/2018	9:22:00	TABLET	10 mg	Oxycodone	REF	1	00901	S12	oral	once	yes		00901	EX	1	S12	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-22T09:22:00	
7	00901-1.512	2018-10-23T18:00:00	23OCT2018	Period-2	11/23/2018	9:22:00	TABLET	10 mg	Oxycodone	YES	1	00901	S12	oral	once	yes		00901	EX	1	S12	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-23T09:22:00	
8	00901-1.513	2018-10-23T18:00:00	23OCT2018	Period-1	10/22/2018	9:24:00	TABLET	10 mg	Oxycodone	YES	1	00901	S13	oral	once	yes		00901	EX	1	S13	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-23T09:24:00	
9	00901-1.513	2018-10-23T18:00:00	23OCT2018	Period-2	11/23/2018	9:24:00	TABLET	10 mg	Oxycodone	REF	1	00901	S13	oral	once	yes		00901	EX	1	S13	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-23T09:24:00	
10	00901-1.514	2018-10-23T18:00:00	23OCT2018	Period-1	10/23/2018	9:26:00	TABLET	10 mg	Oxycodone	REF	1	00901	S14	oral	once	yes		00901	EX	1	S14	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-23T09:26:00	
11	00901-1.514	2018-10-23T18:00:00	23OCT2018	Period-2	11/23/2018	9:26:00	TABLET	10 mg	Oxycodone	YES	1	00901	S14	oral	once	yes		00901	EX	1	S14	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-23T09:26:00	
12	00901-1.515	2018-10-23T18:00:00	23OCT2018	Period-1	10/22/2018	9:28:00	TABLET	10 mg	Oxycodone	YES	1	00901	S15	oral	once	yes		00901	EX	1	S15	YES	10 mg	Oxycodone	ONCE	ORAL	Y	21479	2018-10-23T09:28:00	
13	00901-1.515	2018-10-23T18:00:00	23OCT2018	Period-2	11/23/2018	9:28:00	TABLET	10 mg	Oxycodone	REF	1	00901	S15	oral	once	yes		00901	EX	1	S15	REF	10 mg	Oxycodone	ONCE	ORAL	Y	21510	2018-11-23T09:28:00	

```

PROC SORT DATA=EX2; BY USUBJID EXSTDN; RUN;

```

```

DATA EX4;
LENGTH EXSEQ 8;
SET EX2;
BY USUBJID EXSTDN;
IF FIRST.USUBJID THEN EXSEQ = 1;
ELSE EXSEQ + 1;
KEEP USUBJID EXSTDN EXSEQ;
RUN;

```

```

PROC SORT DATA=EX2; BY USUBJID EXSTDN; RUN;

```

```

DATA FINAL;
MERGE EX2 EX4;
BY USUBJID EXSTDN;
IF FIRST.USUBJID THEN EXSEQ = 1;
ELSE EXSEQ + 1;
KEEP
STUDYID DOMAIN USUBJID EXTRT EXDOSE EXDOSU EXDOSTXT
EXDOSFRM EXDOSFRQ EXROUTE EXFAST EPOCH EXSTDTC EXENDTC
EXSTDY EXENDY EXDUR EXSEQ;
RUN;

```

```

PROC PRINT DATA=FINAL (FIRSTOBS=1 OBS=10);
RUN;

```

CODE	LOG	RESULTS	OUTPUT DATA														
Table: WORK.FINAL View: Column names     Filter: (none)																	
Total rows: 41 Total columns: 17																	
	USUBJID	STUDYID	DOMAIN	EXTRT	EXDOSE	EXDOSU	EXDOSTXT	EXDOSFRQ	EXROUTE	EXFAST	EXSTDTC	EXENDTC	EXSTDY	EXENDY	EXDUR	EXDOSFRM	EXSEQ
1	009/01-1-S1	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:00:00	2018-10-22T09:00:00	2	2	P1D	TABLET	1
2	009/01-1-S1	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:00:00	2018-11-22T09:00:00	33	33	P1D	TABLET	3
3	009/01-1-S10	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:18:00	2018-10-22T09:18:00	1	1	P1D	TABLET	1
4	009/01-1-S11	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:20:00	2018-10-22T09:20:00	1	1	P1D	TABLET	1
5	009/01-1-S11	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:20:00	2018-11-22T09:20:00	32	32	P1D	TABLET	3
6	009/01-1-S12	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:22:00	2018-10-22T09:22:00	1	1	P1D	TABLET	1
7	009/01-1-S12	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:22:00	2018-11-22T09:22:00	32	32	P1D	TABLET	3
8	009/01-1-S13	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:24:00	2018-10-22T09:24:00	1	1	P1D	TABLET	1
9	009/01-1-S13	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:24:00	2018-11-22T09:24:00	32	32	P1D	TABLET	3
10	009/01-1-S14	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:26:00	2018-10-22T09:26:00	1	1	P1D	TABLET	1
11	009/01-1-S14	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:26:00	2018-11-22T09:26:00	32	32	P1D	TABLET	3

```

/* LABEL */
PROC SQL;
CREATE TABLE FINAL_EX AS
SELECT
STUDYID LENGTH=8 LABEL="STUDY IDENTIFIER",
DOMAIN LENGTH=2 LABEL="DOMAIN ABBREVIATION",
USUBJID LENGTH=50 LABEL="UNIQUE SUBJECT IDENTIFIER",
EXTRT LENGTH=200 LABEL="NAME OF TREATMENT",
EXDOSE LENGTH=8 LABEL="DOSE",
EXDOSU LENGTH=40 LABEL="DOSE UNITS",
EXDOSTXT LENGTH=40 LABEL="DOSE DESCRIPTION",
EXDOSFRM LENGTH=80 LABEL="DOSE FORM",
EXDOSFRQ LENGTH=40 LABEL="DOSING FREQUENCY PER INTERVAL",
EXROUTE LENGTH=40 LABEL="ROUTE OF ADMINISTRATION",
EXFAST LENGTH=40 LABEL="FASTING STATUS",
EXSTDTC LENGTH=20 LABEL="START DATE/TIME OF TREATMENT",
EXENDTC LENGTH=20 LABEL="END DATE/TIME OF TREATMENT",
EXSTDY LENGTH=8 LABEL="STUDY DAY OF START OF TREATMENT",
EXENDY LENGTH=8 LABEL="STUDY DAY OF END OF TREATMENT",
EXDUR LENGTH=20 LABEL="DURATION OF TREATMENT",
EXSEQ LENGTH=8 LABEL="SEQUENCE NUMBER"
FROM FINAL;
QUIT;

```

General	Columns	Extended Attributes	Column Extended Attributes		
Column Name	Type	Length	Format	Informat	Label
STUDYID	Char	8			Study Identifier
DOMAIN	Char	2			Domain Abbreviation
USUBJID	Char	50			Unique Subject Identifier
EXTRT	Char	200			Name Of Treatment
EXDOSE	Numeric	8			Dose
EXDOSU	Char	40			Dose Units
EXDOSTXT	Char	40			Dose Description

```
DATA EX_FINAL (LABEL=EXPOSURE);
SET FINAL_EX;
RUN;
```

CODE

LOG

RESULTS

OUTPUT DATA

Table: WORK.EX_FINAL

View: Column names

    Filter: (none)

Total rows: 41 Total columns: 17

	STUDYID	DOMAIN	USUBJID	EXTRT	EXDOSE	EXDOSU	EXDOSTXT	EXDOSFRM	EXDOSFRQ	EXROUTE	EXFAST	EXSTDTC	EXENDTC	EXSTDY	EXENDY	EXDUR	EXSEQ
1	009/01	EX	009/01-1-S1	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:00:00	2018-10-22T09:00:00	2	2	P1D	1
2	009/01	EX	009/01-1-S1	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:00:00	2018-11-22T09:00:00	33	33	P1D	3
3	009/01	EX	009/01-1-S10	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:18:00	2018-10-22T09:18:00	1	1	P1D	1
4	009/01	EX	009/01-1-S11	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:20:00	2018-10-22T09:20:00	1	1	P1D	1
5	009/01	EX	009/01-1-S11	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:20:00	2018-11-22T09:20:00	32	32	P1D	3
6	009/01	EX	009/01-1-S12	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:22:00	2018-10-22T09:22:00	1	1	P1D	1
7	009/01	EX	009/01-1-S12	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:22:00	2018-11-22T09:22:00	32	32	P1D	3
8	009/01	EX	009/01-1-S13	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:24:00	2018-10-22T09:24:00	1	1	P1D	1
9	009/01	EX	009/01-1-S13	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:24:00	2018-11-22T09:24:00	32	32	P1D	3
10	009/01	EX	009/01-1-S14	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:26:00	2018-10-22T09:26:00	1	1	P1D	1
11	009/01	EX	009/01-1-S14	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:26:00	2018-11-22T09:26:00	32	32	P1D	3
12	009/01	EX	009/01-1-S15	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:28:00	2018-10-22T09:28:00	1	1	P1D	1

```
/* XPT OUTPUT */
LIBNAME XPTOUT XPORT
"/HOME/U64168920/PROJECT_WORK_01/EXPOSURE.XPT";
```

```
DATA XPTOUT.EXPOSURE;
SET FINAL_EX;
RUN;
```

```
LIBNAME XPTOUT CLEAR;
```

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      LIBNAME XPTOUT XPORT "/home/u64168920/Project_work_01/EXPOSURE.xpt";
NOTE: Libref XPTOUT was successfully assigned as follows:
      Engine:          XPORT
      Physical Name:   /home/u64168920/Project_work_01/EXPOSURE.xpt
70
71      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
81
```


CODELOGRESULTSOUTPUT DATA

able:XP TOUT EXPOSUREView:Column namesFilter: (none)

Total rows: 41Total columns: 17

	STUDYID	DOMAIN	USUBJID	EXTRT	EXDOSE	EXDOSU	EXDOSTXT	EXDOSFRM	EXDOSFRQ	EXROUTE	EXFAST	EXSTDTC	EXENDTC	EXSTDY	EXENDY	EXDUR	EXSEQ
1	009/01	EX	009/01-1-S1	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:00:00	2018-10-22T09:00:00	2	2	P1D	1
2	009/01	EX	009/01-1-S1	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:00:00	2018-11-22T09:00:00	33	33	P1D	3
3	009/01	EX	009/01-1-S10	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:18:00	2018-10-22T09:18:00	1	1	P1D	1
4	009/01	EX	009/01-1-S11	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:20:00	2018-10-22T09:20:00	1	1	P1D	1
5	009/01	EX	009/01-1-S11	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:20:00	2018-11-22T09:20:00	32	32	P1D	3
6	009/01	EX	009/01-1-S12	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:22:00	2018-10-22T09:22:00	1	1	P1D	1
7	009/01	EX	009/01-1-S12	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:22:00	2018-11-22T09:22:00	32	32	P1D	3
8	009/01	EX	009/01-1-S13	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:24:00	2018-10-22T09:24:00	1	1	P1D	1
9	009/01	EX	009/01-1-S13	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:24:00	2018-11-22T09:24:00	32	32	P1D	3
10	009/01	EX	009/01-1-S14	REF	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-10-22T09:26:00	2018-10-22T09:26:00	1	1	P1D	1
11	009/01	EX	009/01-1-S14	TES	10	mg	Oxybutynin	TABLET	ONCE	ORAL	Y	2018-11-22T09:26:00	2018-11-22T09:26:00	32	32	P1D	3

Obs	USUBJID	STUDYID	DOMAIN	EXTRT	EXDOSE	EXDOSU	EXDOSTXT	EXDOSFRQ	EXROUTE	EXFAST	EXSTDTC	EXENDTC	EXSTDY	EXENDY	EXDUR	EXDOSFRM	EXSEQ
1	009/01-1-S1	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:00:00	2018-10-22T09:00:00	2	2	P1D	TABLET	1
2	009/01-1-S1	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:00:00	2018-11-22T09:00:00	33	33	P1D	TABLET	3
3	009/01-1-S10	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:18:00	2018-10-22T09:18:00	1	1	P1D	TABLET	1
4	009/01-1-S11	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:20:00	2018-10-22T09:20:00	1	1	P1D	TABLET	1
5	009/01-1-S11	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:20:00	2018-11-22T09:20:00	32	32	P1D	TABLET	3
6	009/01-1-S12	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:22:00	2018-10-22T09:22:00	1	1	P1D	TABLET	1
7	009/01-1-S12	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:22:00	2018-11-22T09:22:00	32	32	P1D	TABLET	3
8	009/01-1-S13	009/01	EX	TES	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:24:00	2018-10-22T09:24:00	1	1	P1D	TABLET	1
9	009/01-1-S13	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-11-22T09:24:00	2018-11-22T09:24:00	32	32	P1D	TABLET	3
10	009/01-1-S14	009/01	EX	REF	10	mg	Oxybutynin	ONCE	ORAL	Y	2018-10-22T09:26:00	2018-10-22T09:26:00	1	1	P1D	TABLET	1

FINAL OUTPUT